WILSON

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Solution Architect & Technology Leader — Reusable Platforms and Integration Architecture in a Tier-1 Bank

PROFESSIONAL SUMMARY

Solution architect with 18+ years of designing scalable, reusable platforms and staying accountable for them through delivery and production, across a Tier-1 bank, SaaS products and consumer internet. Began as an engineer and technical lead at Renren, Sohu and Baidu, then held head-of-technology roles (2016–2020) and a project-manager-and-architect role (2020–2022) at start-ups and scale-ups, building 0-to-1 platforms.

Joined UOB in 2022 as a Vice President and leads 25 onshore and offshore engineers on payment and lending platforms. Designed the integration architecture of a payment-initiation hub that connects channels, reference data, vendor systems and core banking, and designs the security and resilience controls.

Designs for reuse, from a customisable multi-tenant SaaS platform to a payment hub that serves several channels. At AI Park, led technical bids that won city projects. Reports status, risks and decisions to senior management and the steering committee, and mentors engineers.

CORE COMPETENCIES

Solution Architecture	Scalable, reusable platform design, multi-tenant SaaS, integration patterns, non-functional requirements (latency, availability, security), technical sign-off through acceptance testing (SIT/UAT) and release
Integration & APIs	Java, Spring Boot and Apache Camel services, API contracts and orchestration, event-driven integration, channel, vendor and third-party integration, hand-off to core banking for settlement
Cloud-Native Architecture	Microservices on Spring Cloud and Kubernetes, API gateway, AWS (ECS, RDS, DynamoDB, SQS/SNS), Kafka, Redis, CI/CD on Jenkins
Governance & Standards	Architecture-review governance and engineering standards (contributed), daily code review and solution co-design, quality gates, secure SDLC, MAS TRM (Singapore technology-risk rules)
Risk & Resilience	Risk and dependency logs (RAID), API and database encryption, credential management, role-based permissions, disaster recovery (DR) and failover drills, 99.9%+ availability, incident root-cause analysis
Stakeholders & Bids	Technical bids, reporting to senior management and the steering committee, mentoring engineers, cross-time-zone teams
Applied AI	Production AI code-review service on headless Claude Code agents, AI document-validation agent, human-in-the-loop governance

SELECTED ACHIEVEMENTS

- **Integration architecture:** Led end-to-end design of a payment-initiation hub — API contracts, orchestration, validation, enrichment and status tracking — integrating digital and branch channels, reference data, and vendor and third-party systems; it hands validated payments to core banking, which executes settlement, and processes the results returned.
- **Multi-tenant platform:** Architected and delivered, from 0 to 1, a customisable multi-tenant SaaS platform on AWS for online-game businesses — KYC, bonus incentives, VIP tiers and multi-channel payments — on Spring Cloud and Kubernetes.
- **Shared platform:** Designed the shared back-end platform architecture (Spring Boot, MySQL cluster, Redis, RocketMQ) for VV's B2B online-office product serving Southeast Asia, governing solution design, cost evaluation, staffing and acceptance.
- **Architecture governance:** Contributes to bank-wide SDLC, coding and code-review standards and to architecture-review governance; reviews core pull requests daily and co-designs solutions with engineers.
- **Technical bids:** At AI Park, led technical bids that won city parking projects in Shijiazhuang, Handan and Beijing, and launched pilots in Shijiazhuang (2016), Handan (2017) and Beijing Chaoyang (2017).
- **Secure by design:** Designed API and database encryption and credential management for payment, lending and customer-data platforms under MAS TRM; personally led DR and failover drills keeping the payment hub and middleware at 99.9%+ availability.
- **Production AI service:** Designed and built an AI code-review service in a regulated bank that runs headless Claude Code agents on each pull request; live since May 2026 after formal architecture and risk approval, it advises but never approves or merges.

PROFESSIONAL EXPERIENCE

United Overseas Bank (UOB) | Singapore — Oct 2022 – Present

Vice President (VP2) / Technical Delivery Manager — Real-Time & Payment Data Services, Group Technology & Operations

Technology leader for two mission-critical platform lines in a Tier-1 bank operating under MAS TRM — cross-border payments and retail unsecured lending — serving millions of accounts.

Architecture & Design

- Personally built the payment-initiation hub's core modules — payment-transaction initiation, concurrency control, the corporate- and user-role permission matrix and processing of payment status results returned by downstream systems.
- Reduced end-to-end transaction latency under high load by tuning thread pools and the Java runtime (JVM) and removing blocking synchronous calls.
- Led the 10-engineer backend team of a regional customer-360 platform that takes daily account summaries from core banking for portfolio analysis; evaluated Cloudera, designed the customer data models and personally built the Spring Batch jobs.

Delivery Oversight

- Planned and architected the phased move of cross-border payments to ISO 20022 (SWIFT's MT-to-MX migration), running old and new message formats side by side: new-format messaging went live in May and August 2025, ahead of SWIFT's November 2025 deadline, and the old path was retired in August 2026 with no service interruption.
- Own scope, roadmap, risks and dependencies, release planning and production rollout with a 15-engineer team, shipping upgrades every 2–3 months.
- Established delivery KPIs (deployment frequency, mean time to recovery, incident trend, change-failure rate) and automated quality gates — a 99% SonarQube coverage gate on all code, regression and penetration testing — with SIT/UAT sign-off on every release.
- Personally prepare the change request for every production release on both platform lines and present the go-live plan, risk controls, rollback plan, test results and database changes to the Change Advisory Board (CAB).

Stakeholders & Production

- Work closely with teams in India, Thailand and Indonesia across time zones.
- Resolve schedule conflicts between the two platform lines by agreeing MVP scope with the requirement, project and technical leads, confirming dependencies and setting a sequence all parties accept.
- Coordinate P1–P3 production incidents end to end — escalation, communication, root-cause analysis and post-incident reports; with no separate SRE team, the platform teams own production support, service-level objectives and Splunk monitoring.
- Lead two platform teams through squad leads and engineering managers, including mentoring; hold decision or joint-decision authority on vendor selection and chair governance of four IT service vendors.

Golden Bridge Drive (SG) Pte. Ltd. | Singapore — Nov 2020 – Jul 2022

Project Manager & Architect

- Designed tiered know-your-customer (KYC) identity verification and rule models for authentication levels and bonus eligibility, triggers and distribution.
- Architected the AWS deployment (API Gateway, ECS, RDS, DynamoDB) and applied data-protection impact assessments (DPIA), API and database encryption and credential management; led data migration.
- Led a 40-person project team, including outsourced staff, introduced Agile/Scrum from scratch and CI on Jenkins and Amazon ECR, and owned delivery budget, capacity and launch readiness.

VV Technology Pte. Ltd. | Singapore — Apr 2019 – Jun 2020

Head of Technical Centre

VV-Life: a data-driven local-life services platform connecting offline merchants in food, fresh produce, travel and everyday services.

- Led VV's 140-person technology organisation through its managers and built its microservices platform from scratch, owning technology selection and the technology budget.
- Designed for performance with Spring Cloud microservices, Kafka-based asynchronous processing and caching that kept load off the database, bringing service response times to the millisecond level.
- Delivered the first VV-Life release in 2019, including a payment subsystem that integrated UnionPay, Visa/Mastercard, Google Pay and GrabPay payments with settlement reconciliation.
- Introduced CI/CD, code standards and containerised deployment with rollback on Kubernetes; defined production-incident priority levels and tiered emergency-response plans.

AI Park | China — Apr 2016 – Sep 2017

VP of Technical Department

- Led architecture, delivery, budget and the technical team for a smart-parking IoT platform connecting car parks to the cloud; introduced a SaaS model aimed at business and government customers.

- Brought Spring Cloud microservices into the architecture, with Kafka for asynchronous flows and caching to shield the database, achieving millisecond-level service processing.

EARLIER EXPERIENCE

- **Hengshunyuan Information Technology — Co-founder (Oct 2017 – Sep 2019, China):** Owned product, architecture and delivery of an audio-learning platform, launching with ~1,500 licensed audio titles and reaching a peak of ~100k daily active users.
- **Baidu — Technical Specialist, Nuomi group-buying e-commerce platform (May 2014 – Dec 2014, China):** As project leader, re-architected the merchant-side order system on Dubbo microservices, delivering two months ahead of plan; designed and coded payment-gateway integration with Alipay, WeChat Pay and UnionPay.
- **Sohu.com — Technical Leader / Project Leader, Sohu News content platform (Mar 2012 – May 2014, China):** Led the re-architecture of Sohu's fifth-generation CMS onto Spring MVC and a Redis cluster, and delivered its OpenAPI messaging platform.
- **Renren Network — Senior R&D Engineer (Jul 2008 – Mar 2012, China):** Developed Renren's shopping website, Renren Mall and CRM, and designed and coded Renren Mall's payment integration with Alipay and UnionPay.

EDUCATION, CERTIFICATIONS & LANGUAGES

- Master's degree in Computer Science and Technology — Liaoning University (2005 – 2008)
- Bachelor's degree in Applied Mathematics — Shenyang University of Technology (2000 – 2004)
- Software Designer (Intermediate) — Qualification Certificate of Computer and Software Technology Proficiency (2006) · PMP — in progress · Site Reliability Engineering (SRE) — formal training
- Languages: English and Mandarin (professional working proficiency in both)